

Testing Plan

1. How To Run

Prerequisites:

Java Version “21.0.7”

java version "21.0.7" 2025-04-15 LTS

Java(TM) SE Runtime Environment (build 21.0.7+8-LTS-245)

Java HotSpot(TM) 64-Bit Server VM (build 21.0.7+8-LTS-245, mixed mode, sharing)

Maven Version 3.9.11

Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)

Maven home: C:\Program Files\apache-maven-3.9.11

Java version: 24, vendor: Oracle Corporation, runtime:

C:\Users\Mayank\OneDrive\Desktop\java\OpenJDK-24

Default locale: en_IN, platform encoding: UTF-8

OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

Windows Powershell

MySQL

Database Configuration

Download the files.

Run the seed scripts on MySQL.

Edit the DataSourceProvider in edu.univ.erp.util to add your username and password for both auth_db and erp_db database configuration.

Eg:

```
HikariConfig cfg = new HikariConfig();
cfg.setJdbcUrl("jdbc:mysql://localhost:3306/auth_db?useSSL=false&serverTimezone=UTC");
cfg.setUsername("//your username");
cfg.setPassword("//your password");
cfg.setMaximumPoolSize(10);
cfg.setMinimumIdle(2);
cfg.setConnectionTimeout(20000);
cfg.setPoolName("AuthHikariPool");
auth_ds = new HikariDataSource(cfg);
```

Compilation & Running*

mvn clean compile exec:java "-Dexec.mainClass=edu.univ.erp.Main"

2. Test Plan

Credentials:

Username:

admin1

inst1

stu1

stu2

Password: "password" for all

Login Window Tests:

1. Appropriate message for incorrect password, non-existent username.
Enter "abc" for username and "password" for password, then "Incorrect username or password" message is shown.
Enter "stu1" for username and "12345" for password, then "Incorrect username or password" message is shown.
2. Appropriate message for empty field input.
Submit without filling one or more fields, then "Please Enter Both Username and Password" message is shown.
3. Dashboard corresponding to role opens on successful login.
Student dashboard for stu1, stu2.
Instructor dashboard for inst1.
Admin dashboard for admin1.
4. Lockout for 5 minutes if 5 incorrect password attempts.
Enter "stu1" for username and any 5 passwords except the word "password" for password 5 times to get locked out for 5 minutes.

Common Tests

1. Logout closes the dashboard and re-opens the login screen.
2. Change Password dialog shows appropriate messages for empty fields, incorrect old password and identical old and new passwords.
Open Change Password dialog for any user, on entering incorrect old password, change is blocked with message "Incorrect Old Password", on entering the same new password as old password, message shows "Old and New Password Cannot Be Same".

3. Maintenance On flag is shown if maintenance is on.
Login as admin and turn maintenance on, then the flag should be visible on the bottom on all dashboards (student, instructor, admin).
4. When maintenance mode is on, all actions which modify the database such as add/drop, adding scores, adding slabs, adding/editing/deleting courses and sections are blocked by showing a suitable pop-up on click.
Login to stu1 or stu2, try to register for/ drop courses, action is blocked by showing message "Maintenance Mode On".
Login to instructor, try to save scores/final grades, action is blocked by showing message "Maintenance Mode On".
Login to admin, try to add/edit/delete courses/sections/users, or assign instructor, or change add drop deadline, action is blocked by showing message "Maintenance Mode On".
5. Student, Instructor and Admin access is called for every database access/change.
6. Fields cannot be empty, number values and dates cannot be nonsensical.

Student Dashboard Tests

1. Course Catalog displays all courses and is sortable by columns.
Login to stu1 or stu2, course catalog should display Introduction to Programming & DSA courses in catalog, sortable by columns.
2. My Sections displays currently enrolled sections.
Login to stu1 or stu2, my sections should display sections for IP & DSA.
3. Timetable displays timetable.
Login to stu1 or stu2, timetable should display section-wise time and room.
4. Register shows course-wise sections for registration.
Should show 1 section each for 1P & DSA.
5. Grades shows component wise grades for current courses and final grades if assigned for completed courses.
Should show component scores for quiz, midterm, endterm for IP & DSA.
Once grades finalised by the instructor, should show letter grades for IP & DSA.
6. Final grades csv can be exported.
Click on export to export csv "my_final_grades.csv".
7. Add drop is blocked if beyond the add drop deadline set by admin.

Try dropping/registering for courses before changing the deadline as admin, it is blocked with a message showing “Add drop period over”.

8. Completed courses are shown and can be exported as csv.
Go to completed courses and export “my_finished_courses.csv”.
9. Duplicate registration and registration in full section is blocked.
Try to register in the same section twice, it is blocked with a message.

Instructor Dashboard Tests

1. Only sections assigned to the instructor appear in My Sections.
Go to admin, and create a new instructor and section, and assign the new instructor to the new section, then see that the new section does not appear for inst1.
2. Can load, add, delete and edit assessments for the selected section from My Sections in the Assessments section.
Create/edit assessments, the same will be visible in gradebook card and to students on login.
3. Can view and assign component wise scores to students of the section and save to the database in the Gradebook tab. Table is sortable by columns and can be exported.
Select section, then edit scores, same is visible to students, and is reflected in statistics and final percentages.
4. Can preview and finalise grades for the selected section in the final grades tab. Sort and export.
Click on preview to calculate preview as per grading slabs, and save to save the scores, after this scores are visible to students and course moves to finished course for student. Verify the same for stu1, stu2.
5. Statistics for course are not nonsensical.
Go to the statistics panel, the values and charts should correspond to the gradebook component scores.

Admin Dashboard Tests

1. Create users, sections and courses.
Create users as per role, nonsensical inputs will be rejected with an appropriate message.
Create courses and sections, nonsensical inputs will be rejected with an appropriate message.
2. Assign instructor from instructor list to a section.

3. Toggle maintenance mode.
Turn maintenance mode on/off, then banner appears/disappears and tasks are blocked/unblocked.
4. Set add drop deadline.
5. Deleting course and sections is blocked if students are enrolled in sections, or sections exist for the course which is being deleted.

3. Test Summary

All the listed tests were passed.

Some system important limitations & choices are mentioned below:

1. It was decided to let scores on assessments be lower and higher than the min and max limit to allow for bonus and negative marking.
2. The weighting rule does not ensure that all weights add up to 100%, again to allow for bonus.
3. Datetime was not implemented for timetables, hence dates and times are not validated for sections, and any resulting clashes of section timings for students or professor is not handled currently.